

6.3 Continuous random variables

Probability and Statistics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

For continuous density f , probability between bounds is area under curve. Normalisation $\int f = 1$ fixes constants; $E(X) = \int xf(x)dx$ and $\text{Var}(X) = \int x^2f(x)dx - E(X)^2$. A cumulative distribution $F(x)$ integrates f and is nondecreasing from 0 to 1.

Worked example: follow the reasoning

For $f(x) = 2x$ on $[0, 1]$, $P(X < 0.5) = 0.25$ and $F(x) = x^2$ inside the interval.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $f(x) = kx^2$ on $0 \leq x \leq 1$, find k and $P(X > 0.5)$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Normalise $k \int_0^1 x^2 dx = k/3 = 1$, so $k=3$ [2]; $P(X>0.5) = 1 - (0.5)^3 = 0.875$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.